

Solution

FRICTION -02

Class 08 - Science

Section A

1. Read the text carefully and answer the questions:

Anita is studying about that how the friction can be increased or decreased by different ways such as by using brake pads in the brake system of bicycles and automobiles. While riding a bicycle, the brake pads do not touch the wheels. But when we press the brake lever, these pads arrest the motion of the rim due to friction. The wheel stops moving. Gymnasts apply some coarse substance on their hands to increase friction for better grip. Powder is sprinkled on the carrom board to reduce friction.



- (i) **(a)** all of these

Explanation: all of these

- (ii) **(b)** Surface of tyres

Explanation: Surface of tyres

- (iii) **(c)** polished marble surface

Explanation: polished marble surface

- (iv) 1. Rough

- (v) **(a)** True

Explanation: True

2. Read the text carefully and answer the questions:

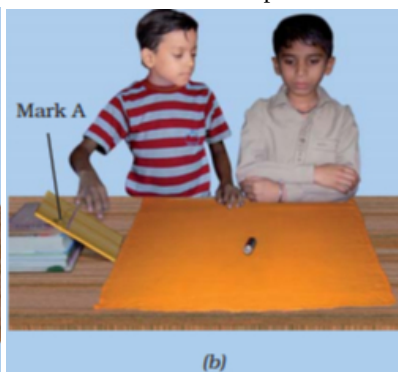
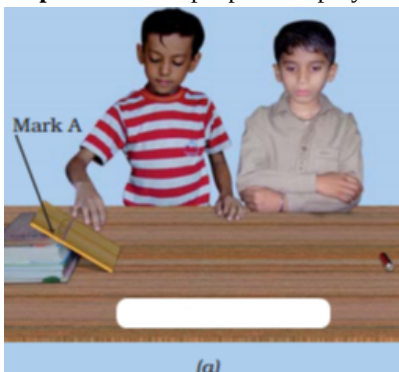
Step 1 - Tie a string around a brick. Pull the brick by a spring balance.

Step 2 - You need to apply some force.

Step 3 - Note down the reading on the spring balance when the brick just begins to move.

Step 4 - It gives you a measure of the force of friction between the surface of the brick and the floor.

Step 5 - Now wrap a piece of polythene around the brick and repeat the activity.



- (i) **(b)** contact force

Explanation: contact force

- (ii) **(d)** opposes the motion

Explanation: opposes the motion

- (iii) **(b)** all of these

Explanation: all of these

- (iv) 1. Spring balance

- (v) **(a)** True

Explanation: True

3. Read the text carefully and answer the questions:

Student X is studying about the friction and relate it with some life experiences that of the outer surface of kulhar is greasy, or has a thin layer of cooking oil on it; would it become easier or more difficult to hold. It is difficult to move on a wet muddy track, or wet marble floor. Friction can also produce heat. Vigorously rub your palms together for a few minutes.



- (i) **(c)** Wearing out due to friction
Explanation: Wearing out due to friction
- (ii) **(d)** Frictional force
Explanation: Frictional force
- (iii) **(b)** Force of friction is very less
Explanation: Force of friction is very less
- (iv) 1. more
- (v) **(a)** True
Explanation: True

4. Read the text carefully and answer the questions:

Step 1 - Take a few pencils which are cylindrical in shape.

Step 2 - Place them parallel to each other on a table.

Step 3 - Place a thick book over it. Now push the book. You observe the pencils rolling as the book moves.

Step 4 - If we use eraser to move the book in this way than to slide it.



- (i) **(c)** greater
Explanation: greater
- (ii) **(a)** Rolling friction
Explanation: Rolling friction
- (iii) **(c)** irrespective of whether the objects are solid, liquid or gaseous.
Explanation: irrespective of whether the objects are solid, liquid or gaseous.
- (iv) 1. less
- (v) **(b)** False
Explanation: False

5. Read the text carefully and answer the questions:

Student X study about the Friction force. He studied about rolling and sliding friction how to increase or decrease friction, how friction is evil for us and how it is useful. Now he wants to study about fluid friction. As air is very light and thin. Yet it exerts frictional force on objects moving through it. Similarly, water and other liquids exert force of friction when objects move through them.

- (i) **(c)** bird
Explanation: bird
- (ii) **(c)** drag
Explanation: drag
- (iii) **(a)** Boats
Explanation: Boats

(iv) 1. Friction

(v) (a) True

Explanation: True

Section B

6. (d) A is false but R is true.
Explanation: Only static friction is a self-adjusting force. This is because the force of static friction is equal and opposite to the applied force (so long as actual motion does not start). So, (A) is an incorrect statement. And Frictional force = μmg i.e. friction depends on mass. So, (R) is a correct statement.
7. (d) A is false but R is true.
Explanation: Friction always acts to oppose any relative motion between surfaces. So, (A) is the incorrect statement. When one object moves over the surface of another object surface of both objects exerts force in a direction opposite to each other. So, (R) is the correct statement.
8. (a) Both A and R are true and R is the correct explanation of A.
Explanation: It is always easier to roll than to slide one body over another. That is the reason it is convenient to pull the luggage fitted with rollers (wheels) to reduce friction. So, (R) is the correct explanation of (A).
9. (a) Both A and R are true and R is the correct explanation of A.
Explanation: Spikes will provide larger frictional force than normal shoes. The spikes will help in preventing the sports players from slipping on the ground. Hence sportsmen use shoes with spikes. Spikes increase the surface of the shoes & make it rough which results in an increase in friction. So, (R) is the correct explanation of (A).
10. (b) Both A and R are true but R is not the correct explanation of A.
Explanation: Objects having streamlined shapes face less friction when moving through fluid than objects which do not have streamlined shapes. So, (A) is the correct statement. The higher the viscosity or thickness of the fluid, the greater will be the frictional force acting on an object moving through it. So, (R) is the correct statement. But (R) is not a correct statement of (A).
11. (b) Both A and R are true but R is not the correct explanation of A.
Explanation: In cycling, the rear wheel moves by the force communicated to it by pedalling while the front wheel moves by itself. So, while pedaling a bicycle, the force exerted by a rear wheel on the ground makes the force of friction act on it in the forward direction. So, (A) is the correct statement. Friction is a contact force. Hence, it is necessary for two bodies to be in contact for the friction to act on them. Therefore, no frictional force acts between the two bodies unless they are in contact with each other. So, amongst these correct options is (B) both assertion and reason are correct but reason is not the correct explanation for assertion
12. (b) Both A and R are true but R is not the correct explanation of A.
Explanation: Friction is the opposing force experienced when two bodies slide over one another. Friction on a surface is caused due to irregularities on that surface. By polishing the surface, these irregularities get evened out and the friction of that surface decreases. So, amongst these correct options is: both assertion and reason are correct but reason is not the correct explanation for assertion.
13. (c) A is true but R is false.
Explanation: Friction is caused by the interlocking of irregularities present on the surfaces in contact. Rough surfaces have larger irregularities while smoother surfaces have lesser irregularities. So, (A) is the correct statement. The frictional force opposes the motion of an object relative to its surface in contact. However, the direction of frictional force may be the same as the direction of motion of the object. So, (R) is false.
14. (a) Both A and R are true and R is the correct explanation of A.
Explanation: Friction force opposes the relative motion between two surfaces in contact. If two surfaces are rubbed with a

small force then simply the opposing force would also be small, but if they are rubbed with a stronger force then friction force will try to oppose the force involved. Hence, it would be higher.

So, (R) is the correct explanation of (A).

15.

(c) A is true but R is false.

Explanation: The frictional force decreases with the decrease in the roughness of the surfaces in contact. So, (A) is the correct statement.

Friction causes loss of energy in the form of heat, sound, light, etc. So, (R) is an incorrect statement.

Section C

16. State True or False:

(i) **(b)** False

Explanation: False; Spring balance is a device used to measure the weight of small bodies.

(ii) **(a)** True

Explanation: True

(iii) **(a)** True

Explanation: True

(iv) **(a)** True

Explanation: True

(v) **(a)** True

Explanation: True

(vi) **(a)** True

Explanation: True

(vii) **(a)** True

Explanation: True

(viii) **(a)** True

Explanation: True

(ix) **(a)** True

Explanation: True

(x) **(a)** True

Explanation: True

(xi) **(b)** False

Explanation: False

(xii) **(b)** False

Explanation: False

(xiii) **(a)** True

Explanation: True

(xiv) **(a)** True

Explanation: True

(xv) **(b)** False

Explanation: False

(xvi) **(a)** True

Explanation: True

(xvii) **(b)** False

Explanation: False

(xviii) **(a)** True

Explanation: True

(xix) **(a)** True

Explanation: True

(xx) (a) True

Explanation: True

Section D

17. Fill in the blanks:

- (i) 1. Smaller
- (ii) 1. Opposes
- (iii) 1. Drag
- (iv) 1. streamlined
- (v) 1. Reduces
- (vi) 1. Opposes
- (vii) 1. Heat
- (viii) 1. Smoothness (or irregularities or nature)
- (ix) 1. lubricants
- (x) 1. Relative motion

Section E

- 18. a. - iii
b. - iv
c. - i
d. - ii
- 19. a. - iii
b. - iv
c. - i
d. - ii

20.	Column A	Column B
	(i) Lubricants	(d) Oil and grease
	(ii) Friction produces	(a) Heat
	(iii) Wheels	(b) Rolling friction
	(iv) Ball bearings	(e) Ceiling fan
	(v) Soapy floor	(c) Less the friction

- 21. a. - iii
b. - iv
c. - i
d. - ii
- 22. a. - iii
b. - iv
c. - i
d. - ii

Section F

23. **Advantages:**

- a. Friction with air makes the windmills work, which are a source of wind energy. Similarly, friction with water makes the turbines works in the hydroelectric power plants.
- b. Friction of propeller blades with air, propels a glider or a small aircraft in air.

Disadvantage

- a. Speed of the object reduces, as fluid friction opposes the motion of the object.
- b. Energy of the object is lost in overcoming fluid the friction when moving through the fluid. This reduces their efficiency.

24. Lubricants are the substances which help to reduce friction. Use of oils, lubricants and greases makes the surface smooth. In machines, friction is reduced by using lubricants. When oil, grease or lubricants is applied between the moving part of a

machine, a thin layer is formed there and moving surfaces do not directly rub against each other. Interlocking of irregularities are avoided to a great extent so that the movement becomes smooth.

25. **Rolling friction** is the force resisting the motion when a body (such as a ball, tire, or wheel) rolls on a surface. **Sliding friction** is the force resisting the motion when a body slides on a surface. The force of friction depends on the area of contact between the two surfaces. As the area of contact is less in the case of rolling than in the case of sliding, rolling friction is less than the sliding friction. That is why suitcases and automobiles are fitted with wheels so that it is easier to move them.
26. Tyres are made by corrugated in order to increase the friction between the tyres and the surface of the road. This increase the opposing force incurred by the tyres. The rougher the tyre is the grip between the road and the road is better. This consequently increase the friction which tends smooth vehicular motion. Hence prevents accidents due to slipping and sliding.
27. a. A small quantity of talcum powder on a wooden surface of carom board reduces friction between sticker and carom board surface. The carom coins slide easily over the board, which is desired.
- b. If we try to hold a glass with soapy hands, it can easily slip from our hands due to no friction. Glass has smooth surface and after putting soap on hands, they also gets smooth. Friction does not work on smooth surfaces due to no irregularities between glass and soapy hands.